The Metaverse
between utopias and dystopias
Horizons and challenges of data protection

Acts of the Conference - January 30, 2023

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This volume collects contributions from
scholars and experts
who participated in the Conference "The Metaverse between
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challenges of data protection," organized by the
Guarantor for the protection of
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for the Protection of Personal Data 2023"

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The Metaverse between Utopias and Dystopias
Horizons and Challenges of Data Protection
INTRODUCTORY REPORT
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As mentioned, it is a true epochal change, in which the digital has played a demiurgic role, not only from an economic, social, and political perspective but even anthropological. To the point that it has become reductive to speak of the digital in terms of technology, when it now represents the backdrop of our existences, a public and private dimension at once, a space of life, a common good.
New technologies have incisively changed the fundamental coordinates on which the relationship between man and the world is articulated, innovating the very concepts of space and time, forms and contents of social relations and, with them, the legal categories as well as the relationship between powers. The very anthropology has profoundly changed, with implications still to be developed and fully understood.
One of the most relevant of these is, presumably, destined to be the metaverse: the next evolution of the internet (recently presented in Davos) and, presumably, more decisive in its effects on the individual and society.
The term is taken from the cyberpunk novel Snow Crash in which Neal Stephenson, in 1992, depicted a dystopian reality dominated by a meta-capitalism with anarchic outcomes, in which the exercise of power

was delegated to multinational corporations, which nearly overwhelmed the social and public function of state systems. The

social role of each individual is, in the novel, defined by their avatar, to whom the privilege of redemption from a life constrained by difficulties and dissatisfaction is entrusted (the protagonist is a delivery person for CosaNostraPizza), with an imaginative existence projected into the virtual. The dimension in which each person's "digital twin" moves is thus, in Snow Crash, all the more utopian the more dystopian the reality that surrounds it, with a torn social fabric, a missing state, and wild private powers.
As in the film Ready Player One, the suggestion of the virtual elsewhere emerges, conceived as having utopian contours, in which to project, sublimating it, the need for redemption from an everyday life.

Quasi Dystopian. As in *Life is Elsewhere* by Milan Kundera, the substitute world—there made of verses—offers the possibility of a second life that sublimates the miseries of the real one, among which is the fact “that people have unlearned freedom.” And even in its hyperbolicity, Stephenson's visionary narrative anticipated many of the aspects (and risks) of platform capitalism that would emerge in the years to come: the gig economy with the danger of algorithmic exploitation, the progressive assertion of private powers, the transposition of life into the digital dimension.

It is significant that the neologism with which Stephenson indicates the virtual reality in which, among the Second Lives of avatars, a computer virus capable of penetrating even the brains of hackers spreads, was taken up by Mark Zuckerberg to identify, already in 2021, the “next frontier in connecting people.” The metaverse is defined by him as an “immersive platform, an embodied Internet that allows one to be inside the experience rather than watching it from the outside,” in which the user does not merely “see” but ends up being the content and not just a cursor on the screen. Even beyond any technological solutionism, the use of the metaverse is hypothesized as an auxiliary function for teaching with interactive online learning or research, with simulations of the effects of designed products.

As futuristic and almost Asimovian as it may seem, McKinsey estimates the value of the business that will be induced by the metaverse by 2030 at $5 trillion, which, according to Meta's estimates, should host a billion people in the next decade. If today there are estimated to be about 11 million headsets worldwide, by 2026 they could reach 50 million units. According to Gartner, by 2026 one in four people will spend at least one hour a day in this virtual (also collective) space, where physical and digital reality converge in an immersive experience capable of even returning sensory perceptions.

This will be a dimension (also articulated on multiple platforms) characterized by persistent, three-dimensional interactivity and thus even more credible, ubiquitous, and transversal, in which it is possible to act and interact through holograms that constitute true digital conduits of the self. The definition of virtual for this new space will further accentuate its etymological meaning of potential, of possibility susceptible to realization. Just as the prefix Meta, not coincidentally taken up by Menlo Park in its rebranding (symbolized by the emblem of infinity), indicates the beyond, the dimension to strive for in overcoming a limit, identifiable and often identified in the physicality of the real.

To the extent that it appears increasingly plausible, capable of simulating reality almost to the point of replacing it, this new dimension (not only spatio-temporal but even experiential and, therefore, existential) could end up representing, at least for many, the place of infinite possibilities, where one can also perhaps free oneself from the immense concreteness of the real. And perhaps its truly atopic nature, as a genuine non-place in the sense of Marc Augé, will not be fully perceived: a space destined for ephemeral transit and not for belonging, detached from social structures and therefore from relationships around which to build a rooting.

Certainly, this multisensory universe with mobile boundaries, covering every sector of life, spanning from work to commerce to entertainment, will propose, but with exponential significance, opportunities as well as risks and issues that emerged with “traditional” internet. This immersive and potentially all-encompassing experience, hyperreal in Baudrillard's sense, will represent the crossroads of some of the most relevant neotechnologies in the context in which we live: A.I., wearable devices, augmented reality, big data, advanced robotics, cloud computing. And if the outcome will be a clear change of perspective, it is not only due to the Hegelian principle according to which quantitative changes can even resolve, if particularly significant, into qualitative distinctions.

According to Matthew Ball, the characteristics of the metaverse are, in particular, scalability, persistence, interoperability, the ability to enable commercial transactions and to guarantee user identity even through avatars, multiple participation from subjects of different natures, transitivity, and convergence between virtual and perceptual experience. The contours that the metaverse will take are not clear, nor which of these characteristics it will possess. We do not know if its structure will be centralized or polycentric, unitary or multipolar (a multiverse?), nor what governance model will inspire it. The future is still to be written. But we know that, despite the variety of forms it may take, the metaverse will have some important implications on at least three aspects, which will also be illustrated by Professors Floridi and Gancitano.

First of all, the transversality and multiplicity of experiences susceptible to realization and the volume of information that can be generated in the metaverse will determine a collection of personal data incomparable to that of the web, in terms of quantity but...
Introduction Report - Pasquale Stanzione

also in terms of quality. Indeed, biometric data will also be included, conveyed, among others, by wearable devices, the abusive use of which must be prevented. The qualitative and quantitative relevance of data flows will lead to a rethinking by design of the consent collection system and the guarantees of transparency in informational obligations. Also, due to the significant interaction rate among users and the consequent need to protect minors from harmful experiences, the guarantee of age verification will be crucial, naturally with systems that do not involve excessively invasive monitoring of user activity.

Decisive will also be the maintenance of the guarantees (now provided by the GDPR, soon by the Artificial Intelligence Act) regarding algorithmic decisions and interactions between humans and machines that simulate human behavior (consider virtual assistants), for which transparency and informed management must be ensured. The technologically neutral approach (and therefore future-proof) of the GDPR can provide a generally complete regulation on the main aspects of this new world, especially thanks to the risk-based approach, which is crucial for modulating protections based on the characteristics of a constantly evolving reality. However, the personalization of content inherent to the metaverse will likely give rise to new demands for protection, in light of new vulnerabilities and even new subjectivities, such as that of the digital twin in which our self will be projected.

Consider categories of data that are entirely peculiar, such as those inferred from online interactions, which may express emotional states, to which a protection proportional to the degree of intimacy revealed must be accorded. Digital creativity will show the urgency of delineating a boundary between data economy (increasingly based on data deduction in the contractual exchange) and monetization of privacy, with all the risks, in terms of freedom and equality, that may arise from it.

The more the immersive and totalizing experience of this "other" life is mediated by the content proposed by algorithms, the more the freedom of the individual from the conditioning exercised by digital tracking must be guaranteed. The metaverse could exponentially amplify the nudging on which the system of filter bubbles is based, inducing conformity and intolerance towards minorities and any expressive subjectivity of difference, with consequent social polarization.

The risk is that the freedom to shape one's own world, promised by the metaverse, is merely apparent and instead conceals a heterodirection of choices induced by microtargeting, whose distorting effects on the formation of individual and public opinion are well expressed by the CA case. In the absence of adequate corrective measures, the guiding capacity inherent in the content selection proposed by algorithms risks indeed becoming, in such a pervasive virtual experience, a true cultural hegemony.

Some significant guarantees will derive from the obligations of transparency and, broadly speaking, accountability introduced by the DSA (particularly for content moderation) and (especially from a commercial perspective, by the DMA), as well as from the enhancement of the guarantees underlying some recent orientations of data protection authorities in the field of targeted advertising.

However, the increase in the power - informational and even performative - of platforms, which will accompany the development of the metaverse, will impose forward-looking choices in terms of governance.

The theoretically possible models are the neoliberal one (destined to inevitably reproduce, due to the heterogenesis of ends, the oligopolistic cartel where each big tech occupies a specific sector); the open model, of an open-source metaverse accessible and modifiable by each user, aimed at promoting broader direct participation but not easily realizable; the publicly managed model, potentially oscillating between the Chinese state-directed model, the South Korean technological autonomism (of which the strategy for a national metaverse is an expression), and a more moderate essential regulatory approach.

Each of these models presents non-negligible risks: that of an even more inequitable extractive capitalism, which ends up leaving the definition of the perimeter of freedoms to the general conditions of contract; that of an anomic management of this space and, respectively, of panoptic surveillance. On the contrary,
for a sustainable development of the metaverse, the protection of economic initiative must be valued, within the limits functional to human dignity and freedom, the participatory instance, and an adequate system of guarantees and responsibilities, on which we will gladly listen to the considerations of President Violante. Regardless of the model that the development of the metaverse will take, the adoption of some essential guarantees is indispensable, aimed at preventing this other dimension, from a utopian space of possibility, from degenerating into an anomic place where rights can be violated with impunity. This is suggested by the very debut of Horizon World, characterized by a true group sexual violence against a researcher, through her avatar. The dematerialization of space and relationships, the transfiguration of the person into a hologram, and the plausible spread of deep fakes can indeed reduce the perception of the (real!) disvalue of the illicit acts committed in the virtual dimension. However, the guarantees to be granted in the metaverse must also extend to the deeper and long-term effects that may arise from what could become a true society of simulation, based on contactless interactions, with its own specific anthropology, which Professors Riva and Benanti will discuss.

In this sense, the psychological impact, also in terms of akrasia and social alienation that the immersive experience of life in another dimension can have, especially on young people, is significant. This dimension is constructed, with a sort of self-deception, according to our desires and modulated on our perceptions. It is not far-fetched to hypothesize a tendency towards disengagement from reality in favor of this elsewhere with dreamlike contours. Furthermore, the implications, in identity terms, of the almost osmotic relationship with one’s digital twin are relevant, with a risk of dependency certainly greater than that of traditional social media. Additionally, the impact that the (already designed) replacement of headsets with a neural interface, capable of projecting this virtual world directly into the brain, will have must also be considered, particularly in the region of the body that is most delicate because it is irreducible to mere biology, as the neural correlate of consciousness. This is a context in which neurotechnologies could, in the not-so-distant future, read thoughts by decoding neural data with brain reading systems. The entry of technology into that inner world where not even the most coercive of powers has ventured can only raise new demands for protection. First and foremost, that internal forum (the sovereign Self, to use Musil's terminology) from whose free formation every other freedom depends, unless we want potentially useful innovations to become the tool for turning man into a non-person, the individual to be trained or classified, normalized or excluded. Data protection today also has the task of preventing this reductionist drift, to promote sustainable and not democratically regressive innovation, especially in light of such an unexplored scenario as that opened by the metaverse. If guided in an anthropocentric direction, with the presbyopic gaze that Europe has had so far on innovation, it can represent that heterotopia capable, like the ship in Foucault, of opening horizons of meaning. And it is to seek this direction, with that courage of responsibility invoked by Hans Jonas as a rule of progress, that we are here today: aware that innovations as profound as those made possible by the metaverse require the most diverse vision and sensitivity. Because we must indeed combine, as David Sassoli suggested, both audacity and wisdom, to avoid the risk of unlearning freedom ourselves. Thank you.

Introductory Report - Pasquale Stanzione

The Metaverse between Utopias and Dystopias
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Luciano Violante
Guido Scorza
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Paolo Benanti

Technique, Power, and Rights in the Metaverse

Luciano Violante

The Metaverse is the most recent milestone of digital transformation and must be interpreted in light of this process. Digital transformation acts in two directions. The first tends to replace intellectual professions. In February, the first test of a robot lawyer will take place in the United States in a location that has not been revealed, for an anonymous defendant. It is only known that he is accused of speeding, a violation that in the United States is often punished with fairly severe penalties. The software will listen from the defendant's smartphone to the formulation of the accusation, evaluate it based on the laws of the State, precedents, and past rulings for similar cases, and advise its client on the best defense through an earpiece. The software has so far memorized laws and rulings from 50 States. The robot is called “DoNotPay,” referring to the high fees of American lawyers. Lawyers are, of course, warned. The ChatGPT - Generative pretrained transformer
- It is a natural language processing tool that uses advanced machine learning algorithms to generate responses and converse similarly to human interaction. It is capable of sustaining conversations with humans and writing texts on any topic. Thanks to the algorithms, the system can respond in a short time and build original results. In this case, the digital replaces the activity of research. The Republic of yesterday, January 29, explained how the young editorial team of the master's program in journalism and multimedia communication at Luiss managed to produce an issue of its periodical "Zeta" through the exclusive use of artificial intelligence, both for the texts and for the images. Reading the entire article is very interesting also to understand how chat can be used and what its limits are today. An interesting article on the risks of this application is published by "La Stampa" today, on page 25. In the replacement of a human activity with a technological function, there is nothing new. The crane, the elevator, the vacuum cleaner do the same thing. The difference lies in the type of activity being replaced. Traditional tools replace physical activity, and therefore primarily concern the body. Digital platforms, on the other hand, primarily replace intellectual activity.

The second direction in which digital technology is moving is characterized by the shift of human relationships and activities into a three-dimensional immaterial environment, which is the Metaverse. The Metaverse creates a three-dimensional environment in which the user can relate to others, buy intangible goods, plots of land, furnish a home, or appropriately dress their avatar, hold a conference or a lesson, dialoguing with those who have entered the same space through the same platform. Access can be made through specific devices, especially headsets and gloves, although in some very recent applications it seems that gloves are no longer necessary because glasses are sufficient to monitor the hand and thus transmit the message. The headsets cost about 1,500 euros. Some platforms have their own cryptocurrency; others, however, require a connection to the user's digital platform. I would advise caution on this point, because then what happens?

Today, the absolutely most frequent use of the Metaverse, about 80%, concerns gaming, and if the player wants to experience an extra thrill, they can buy the Owo t-shirt, costing 25 dollars. Those who wear it experience a slight electric shock, hopefully slight, when their avatar is hit in the virtual war on gaming. Here is a simple example that can clarify the characteristics of the Metaverse: if in the Metaverse I roll a ball on the table, the ball falls to the ground after crossing the edge. In other digital applications, however, the ball continues to go straight. In the Metaverse, therefore, objects know how to move in material space and thus perceive the presence and shape of other objects. The Metaverse is depicted in a cyberpunk novel from 1992 by Neal Stephenson, Snow Crash, where in a polluted and violent society, the only free life becomes that which can be lived in the Metaverse. The hero who achieves the happy ending works as a delivery person for a pizzeria called "CosaNostraPizza," but in the Metaverse, he is a master of the katana.

To grasp the literary precedents of the Metaverse, it is not necessary to resort to cyberpunk. Don Quixote, for example, lives in a meta-reality that he himself has invented, without platforms, through his own imagination. Aldonza Lorenzo, a peasant woman of small virtues, becomes in the novel Dulcinea del Toboso, a beautiful princess to whom Don Quixote promises fidelity, unbeknownst to her. A flock of sheep becomes the Muslim army, and windmills appear to him as enemy giants. His squire, Sancho Panza, tries to bring him back to reality, but Don Quixote remains in his world. He moves from reality into the imaginary world.

A reverse passage from imagination to reality is instead another classic of literature: "Six Characters in Search of an Author" by Pirandello, where from imagination the characters transfer to the stage to ask someone to make them live. Experts, while anticipating great developments of this technology, consider it not yet mature and are cautious because they remember the failure of Second Life, which exploded between 2002/2003, only to then fade away.
Mckinsey published a report on the Metaverse last June titled “Value Creation in the Metaverse,” subtitled “The Real Business of the Virtual World.” On page 10, we read, “Although the concept has been around for decades, today the Metaverse remains difficult to define.” This seems to suggest caution, but on page 37, the report informs us that “investments in 2022 ranged between two and three million dollars. Meanwhile, it is projected that by 2030 they will range between 4 and 5 trillion dollars.” On the other hand, the CEO of Square Enix, a platform for Metaverse games with millions of accesses, argues that it is better not to provide strict definitions so as not to penalize creators.

The Metaverse, like all other digital applications, is presented as virtual, a term that in common usage, as explained by Treccani, indicates something that exists only as a possibility, in contrast to reality. In reality, in my opinion, digital is an immaterial reality because it exists and produces effects. Reality can be material or immaterial; the digital is an immaterial reality.

I allow myself a digression. Until yesterday, the immaterial reality was the sacred. Today, the immaterial reality is also digital. A new term has been in circulation for some time: “phygital,” which refers to the intertwining of what is physical, that is, material, and what is digital, that is, immaterial. This intertwining is now permanent.

Having made this premise, I will now present five brief theses on digital and the Metaverse.

Thesis 1. Digital applications create an environment in which we spend increasingly larger portions of our lives.
Our lives today take place within the permanent interaction with various communication systems, data collection, classification, and authorization based on artificial intelligence. Through digital means, we

inform ourselves, communicate, belong to communities, buy books, do grocery shopping, build and maintain relationships, navigate an unknown city, watch movies, read newspapers, follow soccer matches, listen to music, receive suggestions for cooking, leisure, work, attend or listen to lessons, and we can write to our favorite band or to the person we admire. We appreciate through likes, evaluate the reputation of a person, the sharing of an opinion, and the quality of a photo. This permanent interaction constitutes an environment in which we spend increasingly larger portions of our lives. About 5 billion people in the world are currently connected to the Internet, and more than four billion are active on social media. It is the largest community in the history of humanity living immersed in the same environment. The digital environment, cyberspace, is not territorial; it is meta-territorial. It is neither public nor private; it can cover the territory of all the States in the world, but it escapes each State. Cyberspace, in terms of social, economic, and scientific relationships, as well as emotional ones, is the fourth environment of humanity alongside land, sea, and space. The Metaverse extends cyberspace through the creation of three-dimensional platforms that are similar to each other but each endowed with its own specificities.

Thesis 2. The anthropological change.
Production, like creation, is not a zero-sum process: I produce an object, but the object produces the human, conditions their anthropology, their way of being in the world, their relationship with others and with the external world. This has been the case for electricity, for example, and for the automobile. Ancient products certainly impacted people's lives, but they did not change their fundamental characteristics. The digital environment, on the other hand, is transforming our anthropology.

Since 1997, the year the first iPhone was launched, the forms of communication and information, play, and learning have changed. The relationship between mind, body, culture, nature, and society has changed. A new way of being a human person is emerging, a new anthropology. Some speak of Homo sapiens 2.0: these are the digital natives, those born from 2000 onwards. However, this typology also includes analog immigrants like me, for example, my generation. Analog natives who have converted to digital use, sometimes even in a paroxysmal way. Digital natives and these analog immigrants have short attention spans, frantically tap on their iPhones, send messages, like, and communicate with emoticons. In the naive belief of being freer, this type of sapiens dematerializes social relationships. They send messages instead of talking, express their approval with a like and their disapproval with a frowning emoticon. They measure their own reputation and that of others not based on their own evaluations, but on the number of followers. Every extension of digital possibilities is presented by marketing as an extension of freedom.
Sapiens 2.0
conceives itself as an inexhaustible bundle of rights, without duties even towards itself. The process of dematerialization of social relations, already long underway through social media, has been increased by the Metaverse, with one difference. While the user of social media remains part of the physical material world, the user of the Metaverse moves into the immaterial world created by the platform, through the avatar. The Metaverse has a dehumanizing effect because it disregards touch, which in our civilization is an essential component of relationships with others. Shaking hands, caressing, and hugging are gestures through which humanity has expressed, for millennia, through the body the reaction with others. Corporeality, which is fundamental in the human experience, is sublimated by the technologies of the Metaverse.
Excuse me if I go a bit further. In Christianity, God is not pure spirit, He is not only sight and hearing; He becomes flesh and man through the Son, whose body is wounded and killed. The resurrection following the crucifixion is of the body. To verify that it is indeed him, a disciple touches the body. The same religion assures the resurrection of bodies after the Last Judgment; the body is essential to human civilization.
Today, we are not able to measure all the anthropological consequences of the decorporealization of experience, but it will be appropriate to reflect on it.

Thesis 3. “Homo filmans” and “homo sedens”
In 2000, Giovanni Sartori published “Homo videns.” You will remember that, due to television, Sartori

argued in this book that for the first time in history, the image prevails over the word, changing both communication and the mechanisms of understanding among human beings. The predominance of the image over the word, the scholar continued, would undermine so-called abstract thought and the symbolic activity characteristic of human beings. Thus, the ability to distinguish the apparent from the real and the true from the false would be reduced.
The homo videns becomes increasingly unable to form an idea of his own and reduces his freedom because he absorbs the message that comes to him from television. A recent evolution of the homo videns, this is my hypothesis, is the homo filmans who films everything through the iPhone: the embrace with a personality, a monument, a scene that attracts attention, himself in a particular place, a landscape, an animal.
The homo videns is still, seated in an armchair in front of a screen that he must share with others. The homo filmans, on the other hand, is alone; he is an effect of the digitalization of solitude, the multiplication of solitudes. He does not have to share his screen with anyone and cannot stop. He moves because he must document his mobility, his capacity for observation, and the variety, the interest of the places he finds himself in and the people he meets. In his consciousness, the image makes the moment eternal.
The homo filmans is a narcissist 2.0. He films to attract the attention of those who will receive the images and is therefore forced to astonish. If the filmans witnesses an accident, he does not help the injured but films them. Because being present is more important than providing assistance.
The Metaverse introduces a more modern variant of the homo filmans: the homo distans. The homo distans has a relationship with others through the Metaverse; he is distant. He distances himself from his own body and remains distant from the body of the other, living in a cellophane-wrapped existence. He distances himself from life based on touch and smell, another important characteristic of being human. The experience of the distans is about packaging, about wrapping.

Thesis 4. The deception of disintermediation
In the cyber society, disintermediation is not the cancellation of mediators; it is their replacement. The most dangerous deception is that the old mediators—party, association, church, family, union—presented themselves as such on the public scene, were scalable, and had knowable statutes. The new mediators do not present themselves as such, are not scalable, and do not have visible statutes. Microsoft, Amazon, Google provide us with essential services at acceptable costs and with efficiency, which have become indispensable to us. In exchange, we freely and voluntarily hand over all our data. If the same data were requested by the State, there would be marches and press campaigns.
A reintermediation is underway. The new mediators are the platforms that guide our daily lives, to the extent that...
greater than that of traditional mediators.
I knew the address, the phone number, the executives and the staff of my party, my union, my parish. I could challenge the leadership of the party and the union, I could run for their positions, but not in the parish. Instead, I do not have the address, the phone number of Amazon, nor can I scale it.
The risks are evident.
Microsoft, Google, Amazon, to name the largest platforms, control 64% of the cloud infrastructure market. Microsoft has about 90% of the operating systems for servers and PCs and manages Office, which is the most widely used software package in the world. 92% of email accounts are managed by Microsoft, Apple, and Google.
To further define the weight of digital companies, I recall that in 1990 the top 5 U.S. companies, in terms of market capitalization, were IBM, Exxon, General Electric, AT&T;, Philip Morris; in 2020 they were Apple, Microsoft, Amazon, Alphabet, Facebook, all digital companies.
In the same year, moreover, each of the 5 tech companies was worth more than the combined value of the 76 largest energy companies. We live in an oligopoly. The oligopolists hold in their hands people, businesses, states. If they flipped the switch, the world would stop. This is their power.

The platforms determine public opinion, which is built on the basis of information and has traditionally been shaped by intermediary bodies: parties, unions, schools, churches.

These institutions today are struggling because they cannot withstand social transformations. A void has opened that is now filled, not always positively, by the information provided through social platforms. The flows of thought driven through social media count more than individual intelligences. Those who govern the digital environment have the ability to decide not only what we buy but also what we think and how we orient ourselves in the world.
Technique, power, and rights in the Metaverse - Luciano Violante

Thesis 5. Guarantees in the digital society
An advanced society must protect people.
The digital society relies on an almost paroxysmal intensive use of the network to share information, data, and content. Anonymity, counterfeit identities, and multiple identities are possible, with the ability to disseminate content without being able to trace back to a source that assumes responsibility or can derive legitimate benefit. Anonymity allows for discrimination through hate speech, often with identity characteristics, up to the formation of hate communities, characterized by acts of denigration and aggression, up to digital lynching.
Another side of the coin: anonymity allows less structured individuals, with a more fragile identity, to also be present on the public scene, bringing their positive contribution to social dialogue.
The system must be able to expunge the former while protecting the expressiveness of the latter, without slipping into the ethical state of robots.
The goal is to build a digital civilization capable of giving everyone full awareness of the possibilities, limits, and risks to their freedom of choice. With this expression "digital civilization," I understand the complex of cultural and social aspects produced or conditioned by our society by artificial intelligence. Digital civilization, I emphasize the weight of the noun digital, means a human condition characterized by the autonomy of the individual in the digital environment and by man's dominion over digital technology; the construction of autonomy and the dominion of the person over the algorithm is the goal to be set in the present in order to be free in the future.

The Metaverse between utopias and dystopias. Horizons and challenges of data protection

Life Online

Guido Scorza

In 2006, when the Council of Europe decided to establish the international protection of personal data, it was unimaginable that one day one of these days would be dedicated to reflecting on the impact of the metaverse on the protection of personal data.
And this despite the fact that the word metaverse—which, by the way, Microsoft’s Word autocorrect continues to underline in red—had already been coined several years earlier by the cyberpunk writer Neal Stephenson in his Snow Crash. However, no one at the time could have thought that "metaverse" would one day become one of the most searched words on Google and a phenomenon studied in universities and by authorities around the world.
What we are experiencing, on the other hand, is a season of life in the world without precedent in which, almost systematically, the unpredictable becomes reality.
To convince oneself of this, it is enough to reread a few lines from Homo Deus, a brief history of the future, one of the latest books by Yuval Noah Harari: “For the first time in history, more people die from dietary excesses than from lack of food; death strikes us more often in old age, from old age, than in youth, from infectious diseases; we cease to live more easily by our own hand, through suicide, than due to the risks associated with the presence of soldiers, terrorists, and criminals combined.”
In a handful of years, Harari's prediction, however careful, rigorous, and data-driven, has been literally contradicted by history. The world found itself first fighting against a pandemic and then living through a war strikingly similar to those of the past.
Nothing to be surprised about, therefore, despite its unpredictability in 2006, if today, in Italy, it has been decided to dedicate the...
International Day for the Protection of Personal Data

Towards the Metaverse

Although aware of the fact that today there is no common definition regarding the phenomenon, and there is no certainty that the metaverse has been born to stay with us for long, the current state of uncertainty is well described by recent research indicating that those who believe the metaverse is merely a passing trend are exactly the same percentage of people who think the metaverse is a phenomenon destined to radically change what we currently call the real world. However, it would be a mistake to forget that—whether it is a trend or a serious thing meant to last—the metaverse is already a financially relevant phenomenon that moves billions of dollars worldwide.

But let me try to introduce this session of our work with two provocations revolving around the few certainties I perceive regarding a phenomenon that still seems shrouded in a thick veil of fog and uncertainties.

Let’s Start from the Origins

The entire world discovered the term "metaverse" and consequently began to talk about it only on October 28, 2021, when Mark Zuckerberg, founder and CEO of Facebook, wrote in an open letter: “We are at the beginning of a new chapter for the Internet, and it is also a new chapter for our company… the metaverse will be an immersive platform, an embodied Internet that allows you to be inside the experience rather than just looking at it from the outside.” At the same time, Zuckerberg announced the intention to rename Facebook, his creation, to Meta.

Thus, the metaverse was born in the private sector for the private sector. This, at least, unless we assume that the CEO of one of the corporations that has contributed most to changing the world one day decides to change the famous name of his company and announce a revolution to the world out of pure philanthropy. Whether he wins or loses— as some suggest—Zuckerberg saw and, presumably still sees, despite the hundreds of millions of dollars burned so far chasing his intuition, the metaverse as a lucrative business opportunity.

There is nothing wrong, of course, in this circumstance. Especially since in the following months, many chose to share the bet and invest in the design, development, and management of metaverses or, in any case, immaterial technological realities called the metaverse. That the metaverse was born in the private sector for the private sector seems to me a fact that should suggest some reflection.

And Here is My Provocation

If the Internet, which can be said to have been born in the public for the public—first as a military project and then as an open network for sharing research—has been literally privatized and completely bent to the reasons of the market in a handful of years, the metaverse or metaverses seem destined to be born or, at least, to soon become enormous digital markets of life experiences, the most diverse and heterogeneous.

But if this is the case, then it is plausible that the metaverse or metaverses will be nothing more than a huge private garden or perhaps a sequence of private gardens whose managers will dictate, by contract, the rules and, in this way, decide what billions of people can or cannot do, with good peace of the applicable laws in the countries where the individuals in question live.

In this perspective, it is not implausible that we will witness an increasingly rapid and effective erosion of public powers by private powers. Faced with such a scenario, we must inevitably ask ourselves: is the emergence of the metaverse compatible with a sustainable development of our democracies?

Online Life - Guido Scorza

It is a question, for my part, currently unanswered but one that I believe should be posed on occasions like this.

I Would Also Like to Share with You a Second Provocation

To immerse ourselves in the metaverse, which, despite many uncertainties, is expected to be multidimensional and multisensory as an “augmented” non-place compared to the Internet we know, it will be necessary to share more personal data than we currently do to enter the digital dimension, resulting in at least a quantitative increase in the magnitude of the issues to be addressed in terms of privacy.

And I deliberately say issues and not problems because I believe that, even this turning point of innovation, should be approached with secularism and serenity, rejecting any temptation that suggests we block the transformation of the world and remain in our comfort zone as individuals who have already struggled enough to learn to live in today's digital ecosystem and are not ready for another leap forward, instead focusing on the need to govern, guide, and promote a sustainable development of the metaverse or metaverses to come.

But It Is Not Enough

For what little is known and can be inferred, the business model of the metaverse seems destined to be the same as the Internet we know: simplifying data for services. If this is the case, however, we are on the brink of a systematic, widespread, and penetrating—far more than what we are already experiencing—commodification of personal data.

We will pay increasingly more in data, and the prices will become ever more expensive. It is impossible, in such a scenario, not to ask what the path to govern this drift might be or, rather, what it should be.
and to avert the risk that an uncontrolled commodification of personal data does not lead to a sale of human dignity to the highest bidder and, above all, does not risk making the right to privacy a right solely for the wealthy. For heaven's sake, on this point, it is necessary to be secular and to shun any fundamentalist temptation, but at the same time, one cannot passively accept the idea that personal data is exchanged on global markets like metals or precious stones because data represents us and is nothing more than representations of us in the digital dimension or in the metaverse that is to come. It is essential to identify a path along which contract law—and, under certain conditions, consumer law—can strengthen the right to privacy, ensuring that the individual has greater control over their personal data. This is perhaps one of the most challenging challenges on the horizon.

Online Life - Guido Scorza

How to inhabit digital worlds?

Maura Gancitano

The future has already arrived.
Only it is not yet evenly distributed.
(William Gibson)

The digital reality is something invisible, yet it is having a huge impact on our lives in these years. It seems intangible, inconsistent, yet it provokes emotions, curiosity, addiction, and we spend most of our days in it. That is why it is essential to reflect on what is happening, in order to consciously inhabit the existing spaces and those that will come. In digital environments, we can find ourselves among people very similar to us or very different, and this creates great complexities from which we cannot escape. The digital realm is therefore a real space, but with a cartography that is extremely difficult to define. We do not have maps, yet it grows frantically; we put more and more of ourselves into it, dedicating more and more time to it. The strong feeling is, among other things, that we are somehow compelled to inhabit it and cannot participate in its architecture. We feel the pressing need to talk about it without yet being able to describe a precise perimeter of what it will be, and this is particularly true regarding the Metaverse or, as some argue, the Metaverses. Personally, I have had the opportunity to discuss it at IED Rome regarding creative professions, in the tourism sector at BTO in Florence, and in thoracic oncology at the Feltrinelli Foundation in Milan, to name just three recent examples among countless others. Discussions about the Metaverse are taking place in very different contexts, where it is, however, possible to find the same interest and the same anxieties.

The false that replaces the true

There is a story from 1940 that can help us reflect on the Metaverse: it is “Tlön, Uqbar, Orbis Tertius”

by Jorge Luis Borges. At the beginning of the story, it is reported that Adolfo Bioy Casares, a friend of Borges who had written many stories with him, recalled that one of the heresiarchs of Uqbar had judged mirrors as diabolical. He had read it in an Encyclopedia that contained four additional pages compared to all the other copies of the same edition, which dealt with the world of Tlön. Borges searches for this world and over the years discovers that Tlön is a completely false, invented country, created by a community of biologists, poets, chemists, metaphysicians, engineers, astronomers, painters, moralists, and geometers, not to carry out a parody nor as an artistic or jovial gesture, but to bring into existence a new possible reality, creating a language and a different functioning of things. This group had therefore written an entry in the Encyclopedia, which over time becomes an immense, complete story in forty volumes, telling of a finished and concluded world, with a physics, a biology, a morality, and a mathematics completely different from those accepted by common thought. Reading the recovered fragments, it becomes clear to Borges that Tlön cannot exist and is all fiction, yet he does not stop his search and discovers that the project was born in the seventeenth century, was halted for two hundred years, and found new life and direction in the nineteenth century. In 1944, a reporter from the daily "The American" of Nashville discovers in a library in Memphis the forty volumes of the first...
Encyclopedia of Tlön that has an infinite echo in international print.
Human beings believe in the authenticity of the source, and they begin to replace their own knowledge with fictitious ones: they start to believe in the existence of the planet, schools adopt the primitive language of Tlön and replace all previous knowledge with that of the new Encyclopedia. The story concludes with Borges speaking about himself and the fact that he is at the Hotel de Adroguè, spending quiet days revisiting an uncertain Quevedian translation, which he does not intend to publish, of Browne's *Urn Burial*. It is a story that exemplarily describes the sense of estrangement regarding the changes in the world and an intellectual position that many people hold at this moment even towards the Metaverse. The feeling is that the complexity of knowledge is being lost, along with a twentieth-century intelligence that seems to have become incapable of describing the world. Yet, although it is extremely understandable - it is no coincidence that the philosophical project I founded with Andrea Colamedici in 2015 is called Tlon - perhaps it can no longer represent a comprehensive response to the world we live in.

The unresolved issues of humanity
If we were to read Plato's dialogues again, we would notice that the questions posed by the Athenian philosopher two and a half millennia ago are not so different from those that still traverse our world, even though its morphology has completely changed today. We continue to act in a muscular way, we continue to see nations waging war, we observe that the principle of responsibility developed by Hans Jonas in 1979 can still - as it did then - apply to the possibility of a nuclear conflict. The real virus seems to be the one inherent in human incapacity to coexist peacefully, more than in a specific technological device. This is why we must be careful not to consider the Metaverse, and also social networks, as exclusively bearers of new problems: perhaps these digital environments are rather highlighting all the issues that existed before in the social fabric. The process of vitrification, as Vanni Codeluppi calls it, that is, the process that leads us to put ourselves on display - as if we were constantly in a showcase - began with the transition from small communities to large urban realities, which changed social relationships and self-perception. Since the first industrial revolution, individuals have moved from a social context in which recognition of people occurred in a dimension of mutual trust - even in their diversity - to a world where the Other is a stranger, someone we cannot trust and who does not trust us, but to whom we must try to sell ourselves, that is, to appear appealing.
What has happened in the last four hundred years of history and philosophy has been an essential change in personal identity, in the way we build our days, in the way we understand work. Philosophy has played a fundamental role in this change, spreading ideas - just think of the works of John Locke and Jean-Jacques Rousseau - that have changed the concept of the human person, pushed us to identify with our name and surname, and contributed to the creation of a long series of devices.
2) V. Codeluppi, *Vetrinizzazione. Individui e società in scena*, Bollati Boringhieri, Turin 2021.
3) Andrea Colamedici and Maura Gancitano, *Ma chi me lo fa fare? Come il lavoro ci ha illuso: la fine

dell'incantesimo*, HarperCollins, Milan 2023.
4) Laurent De Sutter, *Per farla finita con se stessi. Antimanuale di crescita personale*, Tlon, Rome 2022.

The Metaverse between utopias and dystopias. Horizons and challenges of data protection
I greatly appreciated the image suggested by Professor Sebastiano Maffettone who, to talk about the new digital environments, spoke of old wine in new barrels: we are witnessing processes that began long before the Metaverses, and it is on these that we perhaps ultimately reflect. In other words, the fear we have is perhaps the fact that the digital environments, which are intangible because we cannot touch them, are environments in which we are actually literally "moving." And it is as if this move represents an extreme way to try again to resolve the human issues that we have not been able to tackle for two and a half millennia.

Generations
Digital worlds, after all, do not appear the same to all people. It depends on characteristics, possibilities, that is, on infinite human variables. And it also depends on the generation to which one belongs. I often find myself talking about the social conditioning related to beauty, to physical appearance, with people from the so-called Generation Z (those born between 1995 and 2010). I usually recount the effects of the means of communication and representation generated between the nineteenth and early twentieth centuries (photography, magazines, large distribution, radio, and then... television) to give an idea of the history that has led us to the current models. When I happen to talk about social networks, I notice that these people see them very differently than I do, as a Millennial (that is, from Generation Y, which refers to those born between 1981 and 1996), and having started browsing the internet at a very young age, I still remember the sound of the modem connecting and a completely different digital structure. My perspective cannot be that of someone who has started watching videos on YouTube since childhood, and I believe this is useful to understand that if we want to comprehend what digital worlds are, what they might become, and how we can inhabit them, we must also encourage the meeting of these different perspectives. Similarly, we cannot exclude the perspective of previous generations, which is often silenced because it seems unable to understand innovations due to being considered too old. On the contrary, we need a multiple vision, and therefore we must ensure that there are more people providing multiple perspectives simultaneously, also to break the prejudices we may have about the generations involved. We have the idea, for example, that young people are the most dependent and the least aware of the risks of the digital world, but I believe that the truth lies in complexity: it is true that there are risks they do not see - for example regarding data protection - but I do not believe that younger generations are entirely naive; rather, they often realize some risks to a greater extent than older generations. It is easier for them to verify information, to avoid falling for online scams, to recognize the falsehood of certain communications much more quickly, and this is also due to a familiarity with devices that even my generation does not possess. Therefore, it is necessary to have a bit of interpretative charity towards all generations and points of view, trying to experience the new digital environments before judging and demonizing them without even having known them.

The Dawn of New Gods
On October 19, 2021, just a few days before Mark Zuckerberg announced the transition from Facebook to Meta, "The Dawn of New Gods," a book I wrote with Andrea Colamedici, was published by Mondadori, whose subtitle was supposed to be "From Plato to the Metaverse."
5) Andrea Colamedici and Maura Gancitano, The Dawn of New Gods. From Plato to Big Data, Mondadori, Milan 2021.

The idea behind it was an analogy between the birth of philosophy, as told by Giorgio Colli, namely that it was born from a loss of certainties and previous Wisdom, and the times we are living in. In particular, the transition from orality to writing seemed very similar to the transition from writing to digital that we are experiencing in these years. It is always a crack to cross, which is leading us towards a very different social, personal, and political configuration compared to the past, and which is somehow leading us to build new gods, that is, new forms of thought that are extremely linked to the

digital world. This "move" is somewhat what Neil Gaiman's novel "American Gods" narrates, in which some gods of the past have managed to survive the twilight, and others entirely new have been generated. The gods are forms that human beings construct to perceive a transcendent sense of things, but also to give a sense to immanence. Since the release of the book, we have started to talk a lot about Metaverses, and we have perceived conflicting emotions: a great unease or a great uncritical enthusiasm, for example from those involved in advertising who see a new territory to explore and through which to earn. Some argue that since we can no longer afford infinite growth in the physical world because our models of production and consumption are unsustainable, having produced the anthropic effect of climate change, in the Metaverses we could, on the contrary, grow infinitely, bringing the same model of production and consumption to that place. After all, what is digital seems intangible and sustainable.
It is a mistaken belief, given that the servers and infrastructures necessary to support digital environments have a significant effect on climate change. On one hand, there is great enthusiasm and a rush because it is a market; on the other hand, there is great unease, but this unease is often an uncritical one, that is, it stems from a lack of experience. I have often found myself discussing these topics in front of people who expressed concern, but when I asked them if they had ever used a VR (virtual reality) headset, I saw only a small number of hands raised. In the spring of 2022, together with Andrea Colamedici at the LabOratorio San Filippo Neri in Bologna, we demonstrated the use of such a device to an audience of several hundred people: a girl on stage wore the headset and two controllers in her hands, while the other people observed on a screen what she was seeing. We administered a program for learning to move, grasp, and throw objects, and it seemed to the audience as if they were watching an infant experiencing physical reality and beginning to understand how to manipulate objects and how to act in the world. The lack of experience needs to be addressed not to change opinions—at the end of the meeting, people were no less skeptical than before, perhaps they had even more questions—but to acquire real awareness.

Digital as Pharmakon
In this regard, the notion of pharmakon can help us, a media term that simultaneously indicates a remedy and a poison. Digital as pharmakon can provide relief, help us overcome barriers and practical difficulties, but at the same time can poison, spoil, and distort our personal identity and social fabric. In the *Timaeus*, Plato reiterates the Hippocratic positions of the school of Cos, specifying that any drug, even beneficial ones, contains within itself both a healing principle and a poisonous one, because it activates an unnatural process and interferes with the order of things, even when it acts to "heal." It seems, in fact, a perfect description of digital technologies. None of us, in the face of such a powerful pharmakon, can predict exactly the effects it will have; what is clear is that being entirely negative apocalyptic—like Borges—or being uncritically enthusiastic while denying the risks of digital technology at the economic level, of national sovereignty, of economic sovereignty—as often happens in business—is still a risk. The idea of pharmakon helps us describe the feeling that what we have in front of us can be both a cure and a poison. This does not mean not taking a position, but rather trying to observe simultaneously ideas that are contrasting but irreducible to one another. That is why it is more necessary than ever to promote and cultivate a contradictory discourse, to ensure that different viewpoints meet, and to always seek to experiment firsthand with these innovations, then confronting them with different perspectives. We have a great opportunity to observe this change and to act collectively in a direction that we deem right and humane. It is necessary to cultivate a debate and avoid the slippery slope fallacy by viewing digital environments as a pure degeneration of physical reality, but to hold together contrasting and irreducible visions.

The Ethical Dimension
My first experience of the Metaverse was during the American elections of 2020: I was reading many articles, following what was happening, and at a certain point, I put on the Oculus VR headset and entered a virtual room where I found myself following CNN's live broadcast with people supporting Biden, Sanders, and Trump who were arguing. It was not like being in a pub in the American countryside, but it was also not like watching CNN live on television or on a computer. It was a different experience, where I could perceive the emotional involvement of those people, without knowing their

physical identities. Digital environments offer us new and multiple possibilities; that is why, in my opinion, simply using the categories of positive or negative represents a limit. There are multiple experiences available to us, but in the end, what remains is truly the dimension of ethics, that is, the dimension of the question regarding what is the right thing to do. I conclude in this regard with a practical example. Currently, there is much discussion about an artificial intelligence, ChatGPT, which stands for Generative Pretrained Transformer. It is a natural language processing tool (or... Natural Language Processing) that uses advanced machine learning algorithms to generate human-like responses within a discourse. When a user inputs a message, ChatGPT processes the input and provides a relevant and coherent response within the conversation. This chat has already been used by one hundred million users worldwide in just a few weeks and promises to accelerate many processes, such as those involved in text processing. Being an imperfect version, it makes many mistakes, carries numerous cognitive biases, and sometimes cites non-existent scientific studies as sources. Let us consider a student who decides to use ChatGPT to write a paper for their university professor, to do it quickly and avoid having to do it alone, to deceive the teacher and gain an advantage over other students, thus citing unverified sources. Who will be responsible for that work, the chatbot or the student? A simple example to illustrate that the distinction always lies in human action. Ultimately, what we should question is the ethical dimension: what makes us agents, that is, what allows us to act in the world in a conscious and responsible manner.

How to inhabit digital worlds? - Maura Gancitano

Anthropology of the Metaverse

Agostino Ghiglia

Talking about the metaverse today is somewhat akin to discussing the internet in the 1970s. Not all predictions about what it would become have proven true. Yet, the internet has led to a series of revolutionary technological innovations, such as the ability for computers to communicate with each other over long distances or to connect from one web page to another. Innovations that have subsequently formed the building blocks for creating the “abstract structures” through which we navigate online: websites, apps, social networks. So what is the metaverse, or rather, what are the metaverses today? What developments can we expect? The most appropriate answer is that metaverses do not exist today, and I do not know if they will exist. They represent a suggestion, an application of immersive augmented reality, a seductive commercial idea light-years away, in its positive appeal, from the dystopian and surrealistically mercantilistic world created by Stephenson in “Snow Crash.” Some might object: “But how? Don’t they already exist?” citing, as examples, World of Warcraft, a persistent virtual world where players can buy and sell goods; Fortnite, which offers virtual experiences such as concerts and exhibitions. One could also counter that there are 141 existing virtual worlds, of which 62 are already freely accessible, persistent (i.e., existing independently of the presence or absence of a subject), economically active, equipped with 3D graphics, and interoperable. Is all of this really the meaning of the metaverse? Is it a new type of video game enhanced with a headset and two advanced joysticks? Or the possibility of exchanging NFTs or purchasing virtual movable or immovable goods? In my opinion, metaverses are currently a vision of what the future could be, in which our personal and commercial lives would be conducted digitally, in parallel (or as a substitute) with our lives in the physical world. In light of this premise, it is legitimate to ask whether the Big Tech giants are, in fact, guiding - or pushing - us towards Zuckerberg's prophecy of a new “operating system” that aims to be the foundation of our existence; that is, towards the only admitted reality: the virtual one. Who knows if, as we distance ourselves from our residence on Earth, this “non-reality” also reveals an eternal virtual reality (in Snow Crash, demon avatars dragged deceased avatars into underground galleries).

In the face of this potential scenario, and by some commercially “progressive dismission” of the human dimension - where digitalization derealizes and disembodies the world - I believe it is urgent to develop a more critical perspective regarding neo-dogmatic and overly simplistic praises of the Metaverses; praises that obscure the potential risk of this post-human dystopia: namely, the black hole of the online that engulfs offline reality, life as such. As a consequence of this migration of the mind towards the

virtual realm, the real world becomes more insubstantial. Information fades the stable outlines of objects and, with their fleetingness, leaves us without the anchor of tangible contact with reality. Even today, the gradual immersion in virtual reality (which precedes the idea of the metaverse by almost twenty years) risks making us blind to the silent, unremarkable things, to those, so to speak, excessively ordinary and habitual yet vivid and real. We must not, of course, think of ourselves in this room.
the existing legal frameworks) and the moment in which such regulation is actually established.

1. The digital acceleration, in the virtual realm, risks leading us towards a progressive dimension of bodily disempowerment, of functional illiteracy taken to the extreme. On the other hand, how many of us, under 40, are still able to write with a handwriting that would allow for a passing grade in elementary school? Just as the informational chaos has thrown our cognitive system into disquiet (we are increasingly targeted by information that we are unable to discern or select), so augmented virtuality could render part of our sensory and existential system crippled.

2. As beneficial as technology has been (and continues to be) in its countless applications for human progress and the improvement of our lives, we must never forget that at the center there is and must always be the human being. This may seem trivial, but it is not to be taken for granted if we emphasize the concept of human being without replacing it with that of user, consumer, or operator.

3. If we share the premise that technology is, therefore, a tool of our own creation, the prosthetic extension of the state of the human soul within the matrix of existence, it seems that the human being, due to an overwhelming technological-virtual immersion, risks losing their power to act, their decision-making autonomy, their free will in the face of a new world that is becoming increasingly complex, where the gap between reality and truth widens hour by hour.

4. To return to the metaverses, who is deciding, for example, to lead us, hand in hand and with our consent, into them? And in what way? Certainly through cultural influence, targeted information, and the curiosity of the novelty that everyone talks about in an inversely proportional manner to the knowledge of it: by capturing the imagination, the ground is prepared—modifying perceptions and values—for the new vital and legal model that is intended to be "suggested."

5. Since the dawn of time, there have been very few cases in which the law has preemptively regulated specific human behaviors that have not yet occurred but are only perceivable on the horizon. Generally, normative production is the consequent effect of a triggering cause constituted by the emergence of new behaviors that need to be regulated positively or negatively.

6. In this scenario, however, the law—regardless of the metaverses but simply due to the hypersonic advance of the digital age—should strive to keep pace with it, and ethics should instead arrive before legal regulation. Ethics, in fact, must be able to read the fundamental anthropological grounds, that is, those concerning the common traits of societies, to the point of being unavoidable: education, work, care, etc. These are themes where technological acceleration and potential incidents continuously and obsessively chase each other.

7. Not to mention the necessity, on the part of ethics, to understand social mutations by educating against reasoning in stereotyped ideal types or those induced for commercial purposes. In the digital age, in short, technology must continue its process of incessant development and improvement in accordance with its own times, while the law must strive to reduce the distances between the moment when the necessity to provide specific regulation for new phenomena arises (to which existing legal frameworks cannot be applied) and the moment when such regulation is actually established.
analogously to old normative precepts) and that in which the legislator truly acts. All this is to prevent the major players in the technological market from self-establishing, amid inertia, indifference, and delays of legislators, the terms for their commercial interests, creating a new law and a new system of life for their own use and consumption. The emerging potentials of the metaverses are predominantly owned and developed by their creators. Such companies seek to manufacture the perception that the metaverse and its countless offshoots must be a new and futuristic world and, above all, that there is a need for it. But will the stories of the metaverse be real? Will they be ours? Or will it just be a game?

Anthropology of the Metaverse - Agostino Ghiglia

As I come to a close, the evolution of the "new parallel universe" and its potential risks and interests must find a safeguard in a thorough and constant interdisciplinary interaction that cannot be limited to the legal sector alone but must involve interaction with those of ethics, philosophy, and science. Consequently, serious reflections can mature on how much of the "virtual" individuals (not the technological giants) can afford to advance and prosper; to promote the development of a society of participation, making the best use of innovation and technological progress and the resulting economic and social effects. The idea that an open metaverse is possible, designed by design to respect the principles of security, transparency, minimization, and legitimacy of data processing, safeguarding the protection of digital rights and the fundamental values of dignity, autonomy, and freedom, does not seem easily reconcilable, at the moment, with the plans of the technology industry, engaged in massive investments and sophisticated business plans to seize the governance of the virtual world; plans and investments potentially heralding power asymmetries, technological vulnerabilities, exclusion costs, and abusive discrimination.

Returning to the main theme of this European day on the protection of personal data, I believe that technologies, by their very nature, are neutral and are not truly dystopian or utopian. However, the people, the companies, and the immeasurable interests underlying them could be. Therefore, for the metaverse, if it exists, not to be controlled and led solely by the interests of huge commercial enterprises, we must design it for human beings, not for users. The largest technology companies are investing enormous sums of money in the creation of ultraverses whose main characteristic will be the fusion between the virtual world and the physical one.
The Metaverse between Utopias and Dystopias. Horizons and Challenges of Data Protection

Allowing this possibility is one of the less obvious and painless consequences: the metaverses—in their multiple manifestations—aim to function like our minds and not just through them. This capability makes the metaverse itself an incomparably different technology compared to those that preceded it. While television and social media are persuasive technologies, due to their ability to influence people's attitudes and behaviors, the metaverse is instead a transformative technology, capable of altering what people think reality is. To achieve this goal, metaverse technologies hack various key cognitive mechanisms: the experience of being in a place and in a body, the processes of brain-to-brain tuning and synchrony, and the ability to experience and induce emotions. Clearly, these possibilities define entirely new scenarios with both positive and negative outcomes. Educating ourselves about the challenges that the invoked "new world" might present is a necessity. This requires a "human," integrated, and multidisciplinary approach with all stakeholders, starting with states at the supranational level. The metaverse that, thirty years ago, science fiction writer Neal Stephenson depicted in "Snow Crash" was an escape route into an alternative virtual world from the hellish landscape of 21st-century Los Angeles. The main character of the novel, Hiro Protagonist, lived in a container, worked as a pizza delivery driver and as an agent of a sort of post-modern CIA, and fought (even with his real katana) in a brutal anarcho-capitalist world, marred by threatening monopolists, in the throes of an irreversible economic collapse and environmental degradation. But that is just science fiction… Therefore, we must work from now on to ensure that science fiction remains such and to use technology to improve "reality" and not just and necessarily to "augment" it by modifying it, continuing to foster a REAL connection with people and the world around us. This is what we humans were born for, the result of two million years of evolution. Some might argue that we should then fear and curb technological development and return to a simpler way of life. But nostalgia and fear of evolution, of the future, of progress cannot be the answer in a sort of Platonic "myth of the cave" in reverse. Technology must be used to optimize human experiences, to help us live the physical life better, not to replace it…
Neurosciences of the Metaverse

Giuseppe Riva

I would like to thank Agostino Ghiglia for this long but passionate introduction, which introduces a critical theme that is the anthropology of the Metaverse. I am a psychologist and I will share my reflections from two perspectives. The Italian perspective, from Milan at the Catholic University where I am the director of the Humane Technology Lab, a laboratory aimed at analyzing the impact of technology on human experience. The international perspective, as I am also the president of the International Association of Cyberpsychology, a group of researchers—250 worldwide—who are working on the Metaverse to try to understand what it is and what it will be.

Because, in fact, what the Metaverse will be, we still do not know. I have prepared some slides because speaking remotely is certainly less effective than speaking in front of an audience, and in this way, I hope to share with you a series of reflections that I hope will be interesting.
Neurosciences of the Metaverse - Giuseppe Riva

What is the Metaverse from a psychological perspective? From a psychological point of view, the simplest way to describe the Metaverse, beyond what we have seen so far, is to describe it as an evolution of the web. We move from the 2D web "I see from the outside"—the classic one we are used to on a computer screen—to a 3D web in which "I am inside." At the same time, from a static web—like the one we are used to seeing daily—we enter a dynamic and permanent web.

In particular, this permanence leads physical reality to merge experientially with digital reality. It is the experience of the "phygital" that Pasquale Stanzione mentioned at the beginning, which I believe is indeed the true novelty of the Metaverse alongside the sense of presence. In recent months, I have visited several laboratories around the world, and indeed, what strikes most those who experience the Metaverse is precisely the fact of "being inside." I am no longer a spectator, but I am within a digital experience. It is something truly difficult to imagine if you do not try it, but indeed it is so.

But what if the Metaverse were a flop? The truth is that today the Metaverse does not really exist. Several speakers have mentioned this, but indeed having tried what is available in laboratories, and the technologies that are currently available, what we see is only a pale reflection of the Metaverse that will be.

It reminds me a bit of the world of television in the 1950s. In the 1950s, the dominant medium was cinema, and no one thought that watching something on television—a burgeoning medium—on a small screen—17 inches—would make sense. A television cost quite a bit of money, there was no content, there were no programs, there were no channels... And yet we know how it ended.

What made the difference was the creation of new content and a significant enhancement of technological development.
Neurosciences of the Metaverse - Giuseppe Riva

Indeed, what I have seen in these months is that the amount of investments and laboratories working on the Metaverse is beginning to be substantial. We know, as we have said, about the work of Meta. But it's not just Meta… Apple has over 1000 people currently working on its virtual reality headset. Microsoft has about 1800 people working to integrate the Office suite into the Metaverse. These are enormous figures. All the data we have available tells us that by 2025, investments could reach 100 billion dollars. Keep in mind that to send a man to the moon, NASA spent 26 billion dollars in current figures. Therefore, it is inevitable that the Metaverse will exist. It is just a matter of time. The problem, however, is: why so many investments in the Metaverse?

From a psychological perspective, what is rarely mentioned—but is indeed one of the explanations for understanding what the Metaverse will be—is that the Metaverse functions similarly to our mind. When we think about how our mind works, usually, the metaphor we use is that of a computer. In reality, cognitive sciences have changed the metaphor for about a decade. What cognitive sciences tell us is that our mind is a simulation machine that tries to predict threats and what will happen before it can occur, in order to survive in an increasingly complex environment.

In other words, what our brain does is construct a simulation of the body and the surrounding space. This capacity for simulation is what drives our intentions and our behavior in the world. This capacity for simulation is indeed what has allowed humans to always be one step ahead, to imagine, to dream. The ability to have intentions arises precisely from the ability to see not only what is, but what will be.

And these are not just dreams, but the ability to plan and organize one’s destiny.
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The Metaverse works more or less in the same way. From an organizational point of view, the Metaverse is indeed a system that seeks to predict the sensory consequences of the user's movements (what the user does) and tries to reproduce the experience using the power of technology. The more the model of virtual reality and augmented reality resembles the model that the brain uses to describe reality, the more the subject feels present within the world of virtual reality. This is the great power of the Metaverse, namely the ability to deceive our mind.

Earlier, Sebastiano Maffettone spoke about the dreamlike capacity of the Metaverse. Indeed, we can say that it is so. Just as a dream is capable of reproducing simulation mechanisms that the subject uses in reality, so does the Metaverse.

But why then use the Metaverse? One of the great advantages of the Metaverse, from a marketing and advertising perspective, is that if the Metaverse functions like our mind, analyzing the Metaverse makes it very easy to understand who we are.

From the analysis of the numerous available studies, it clearly emerges that by analyzing the verbal and non-verbal behavior of the user and their behavior within the Metaverse, we can know everything about them. We can understand their personality, personal tastes, and the emotions they experience.

Therefore, it is obvious that, from the point of view of analytical and control capacity, the Metaverse pushes the potential for analysis of social media to the extreme. Social media already provided a series of relevant information about subjects; now, in the Metaverse, the subject even becomes transparent. Simply by seeing what the subject does, without needing to do anything else, by observing their eye movements, not even the movements of the body or the avatar, I can understand who they are and what they do. This clearly requires protective mechanisms. Because whose information is it in the Metaverse at the moment I behave and act within this new digital environment?

The other element that the Metaverse clearly offers is a persuasive system, unlike any that has existed before. Social media have made fake news possible, which are already capable of modifying user behavior.

Through the Metaverse, we can go further: create fake news in which we can be present. Fake news are no longer just news but experiences, and these experiences can significantly modify our attitudes and emotions.

Thus, the Metaverse has two great risks: the ability to make all the characteristics of the user—personality, values, interests—visible to the companies operating in the Metaverse, and a persuasive power of dimensions never seen before in the world of technology. As Agostino Ghiglia said earlier: “From the persuasive dimension, we are almost at a transformative dimension,” meaning we can make the subject capable of believing everything they see. Because what they see, they are experiencing directly.

However, I am not only negative in relation to the Metaverse. I believe that the Metaverse can also be very useful in trying to remedy one of the great problems of social media today, namely the transformation of the sense of community.

In recent years, I have studied a lot about what makes a community. And indeed, what we know from neuroscience studies is that the central characteristics of physical communities that have allowed human evolution (the community is born and develops within a place through the relationship between community members, who progressively come closer and work together) are clearly being called into question by the spread of social media, with very strong anthropological effects.

We can assert that, on one hand, places are a fundamental element of our identity, and neuroscience has always emphasized this. In particular, we have specific neurons, GPS neurons, that activate when we are in places. And these places define our identity, so my identity is defined by the places I frequent: I am a professor because I am at the university, I am a worker because I am in the office.

When places are absent, I feel without a part of myself.

We experienced this when, during the pandemic, we had to stay home, and the lack of places made us feel somewhat devoid of identity.

Physical communities then allow people within them to come closer, to reduce distances. We know that one of the biggest problems of social media today is precisely the difficulty of confrontation. And this is facilitated in physical communities because there are particular neurons, mirror neurons, that help us understand the emotions that the other person feels, and by experiencing the same emotions as the other person in front of us, obviously, our approach towards them changes.
The last great power of physical communities is that they allow synchronization among multiple minds. We know that in the world of computers, parallel computing significantly enhances the capabilities of the computer. This also happens in physical groups. A series of very recent studies—using a technique called hyperscanning—has allowed for the verification of the presence of a synchronization process of brain waves that enables the group to go beyond the sum of its parts. In practice, being in a physical community where people work together produces a creative effect and enhances social dynamics that transform the group into something more than the sum of its parts. All of this does not exist in the digital world. Today, social media have completely removed the place and interactions in physical locations from the sociality of younger people, with very important anthropological effects. Here projected on the screen are two articles in which I have detailed what I am telling you now.

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So what happens in practice? Since places are no longer the aggregating tools in online communities, how do young people meet? Today, young people find themselves not in physical communities, but in "communities of practice." Communities of practice are groups of individuals who come together not because there is a place that acts as an aggregator, but because they have a common goal. This common goal drives them to be together and to try to pool their resources to achieve what they want: we want to play a video game, we all meet to play video games. We want to talk about fashion, we meet to talk about fashion, etc.

The Metaverse between utopias and dystopias. Horizons and challenges of data protection

However, this represents a substantial difference from an anthropological point of view. While physical communities are communities of diversity because the sense of place and the capabilities of our minds progressively allow for the reduction of distances, fostering integration and confrontation (which are at the basis of democracy precisely because in physical communities there are different thoughts and at this point a regulatory mechanism is needed that allows everyone to express themselves), digital communities are communities of equals because what aggregates them is precisely the fact of having a common goal.

But what are the problems? The first is that communities of practice are not interested in confrontation; there is no need for democracy because everyone thinks the same way. And then, once the common goal is achieved, the community no longer exists. The aggregating dimension—the community—works only until the common goal is reached; then everyone is free. The result is paradoxical: despite people being together, in the end, they feel alone.

From this perspective, the Metaverse could be different. The Metaverse could bring back places and communities within the digital world.

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A whole series of more recent studies have demonstrated that the mechanisms we have just seen, which are typical of physical communities, are also present in the Metaverse. For this reason, by using the Metaverse, it could be possible to restore to those (digital) communities the confrontation and dialogue that are currently lacking within social media. It is no coincidence that today Meta believes that the killer application of the Metaverse is its use to eliminate the physical office. With the pandemic, we have seen that smart working is a tool, not the solution. Precisely because communities cannot be formed on Zoom and Teams, with all the related problems. The Metaverse could overcome these issues while simultaneously allowing for the cost and time savings of smart working that companies and workers are currently reluctant to give up.

The other major area where the Metaverse will likely have an impact, and where work will be needed, is in education. For this reason, my university, the Catholic University, has just launched an experiment—the “Metaversity” project—because the Metaverse could succeed in improving distance learning, which is currently even less satisfactory than smart working.

Meta-ethics or synthetic living

Paolo Benanti

[…] The ethics of the Metaverse is a very important and complex topic. In general, it can be said that the ethics of the Metaverse focuses on how to ensure that interactions and experiences within the Metaverse are safe, fair, and respectful of human rights. There are many ethical issues to consider, such as privacy, intellectual property, character representation, and user safety. Another important issue is that of responsibility. Who is responsible if something goes wrong within the Metaverse? How can users' rights be guaranteed, and how can they be protected from violations of their rights? Furthermore, the ethics of the Metaverse must also consider how the Metaverse will influence society and culture in general. For example, how will the Metaverse influence social relationships and the economy? How can we ensure that the Metaverse does not perpetuate existing discrimination and inequality in the real world? In general, it is important to continue discussing and...
reflect on the ethics of the metaverse as it becomes increasingly important and widespread in society. It is essential that there is a continuous dialogue between developers, users, and society at large to ensure that the metaverse is a safe, fair, and respectful place for human rights for all. This brief abstract may seem, in perhaps not a brilliant way, to cover the highlights of an ethical-anthropological discourse on the challenges of the metaverse. However, due to the way it has been composed, it is able to reveal, perhaps, the anthropological essence of the challenges posed by the metaverse: the text is not actually elaborated by a human but is the product of a generative artificial intelligence. The metaverse thus becomes the realization of a synthetic "world." Analyzing this new season of the synthetic is the purpose of my contribution, which, being placed at the end of an intense morning rich with many authoritative contributions, will take the liberty of directing the reflection towards a horizon of intellectual provocation.

Consider the synthetic diamond, indistinguishable from a real diamond except for two details: by law, it must have a serial number, engraved with a laser and not visible to the naked eye, inside it, and it is devoid of any imperfections.

An unavoidable question then arises for philosophy, the human sciences, and theology: are we heading towards a reality in which the distinction between natural and artificial is destined to disappear? If so, what will be the consequences of this new understanding of reality? And what perspectives will open up? If the technological creation of the synthetic frees us from some limits that have characterized our history, it also deprives us of a category that had helped us delineate boundaries and clear lines between what is lawful and what is unlawful. If we have become demiurges, if we are capable of creating matter, energy, and now even life and new worlds, the question becomes increasingly urgent: can we, ethically, do everything we are technically able to do? If the nature of the thing we decide upon no longer helps us, as the realities we are discussing are synthetic and not natural, we need something that goes beyond nature. This statement may seem, and is indeed, a request for an adequate metaphysics for understanding synthetic reality. However, pop culture is the main actor in the described processes, and it does not wait for the development and dissemination of adequate philosophical explanations. Pop culture has already developed a synthetic culture that, in its own way, interprets and justifies synthetic reality but does not investigate its nature. The absence of indications of a natural kind, that is, related to the being of the thing in itself, does not leave an unreal void but causes the current mass culture, what we might define as the cultural mainstream, to produce indications not in a philosophical manner but in a mythical way: mass culture is producing myths that orient and justify the creations and use of synthetic reality. In particular, it is the posthuman and transhuman movements that act as mediators of this modality, exploiting the same mass communication means that generate pop culture.

The posthuman movement, heralding the anxieties, hopes, and fears of our time, proposes a transformation of the foundations and modalities of life so radical that it speaks of anthropological mutation. It is not technology in itself, understood as an autonomous agent, that can provoke this mutation: it is not possible to understand the posthuman movement by reading the technological landscape separately from the cultural one because both influence and determine each other. There is a place in pop culture, in our view, where this intersection and mutual interdependence is perceived and understood better than elsewhere. One must look at that literary genre that emerged in the twentieth century, which has gathered both the cultural ferment most sensitive to technological evolution and the fears and dreams associated with it: science fiction. Through the science fiction narratives of the twentieth century, it is possible to trace those visions of the human and its relationship to the world of technology that are characteristic of the posthuman and that justify and shape synthetic reality. In the volume Mythologies from 1956, Roland Barthes made it possible and thinkable to systematically study modern mass culture. In his work, with particular acuity, examining the unconscious semiotic aspects of pop culture, he opens the doors to the applications of semiology and semiotics, identifying those phenomena of signification in everyday life that characterize an increasingly consumerist and materialistic society.

The basic hypothesis of Barthes is that pop culture has a significant emotional impact compared to other types of culture because it is capable of recycling classical myths, transforming them into new forms of representation that, in turn, fit into the mindset of the modern world. Thus, synthetic culture, the essence of this pop culture, is a

mythological system that recycles the classical system of original myths, and it is within this "secondary" system that modern spectators develop their ethical and moral models. In the past, these were established through religious-philosophical systems, later transformed into norms of ethical behavior. In mass cultures, however, the models are ephemeral and disposable, just like the material and cultural products that must always be recycled in order to be economically sustainable. Ethical problems are represented in this context, for example, in television programs and included in narrative texts primarily aimed at entertainment. The new mythology thus constitutes a superficial philosophical language. We can summarize this path as a transition in science fiction literature from the figure of the monster to the protagonism of the cyborg, a proto-form of synthetic man. The monster, to some extent, until the first half of the twentieth century, indicated what was an established order, a will that man perceived as preceding his own will and that was to be respected. After the mid-twentieth century, the cyborg emerges in the popular imagination of the twentieth century as an ambiguous and unsettling figure: certainly linked to technological developments without which it could hardly have been conceived and presented on the pages of science fiction books, on cinema screens, in comic strips. The cyborg is, in good approximation, a cultural incarnation of synthetic reality. The cyborg was born in the 1960s in a context where it seemed that humanity's technological challenge was the imminent conquest of space, yet the characteristics that Clynes and Kline attribute to their cyborg seem destined for the creation of individuals capable of surviving a nuclear war and the subsequent atomic winter.

The cyborg, paraphrasing Caronia, carries within itself the fears and anxieties of its time, which are perceived as a transformation of the foundations and modalities of associated life so radical that more than one speaks of "anthropological mutation." It is not technology understood as an autonomous agent, of course, that provokes this mutation. Technology is the offspring of human activity, and as such, it is not a cause but a striking symptom, a mediating element, a symbol of the transformation that envelops us. This transformation of our culture is compared to that which was experienced in the late Middle Ages, according to an analogy that has circulated for some time in Western culture and of which Umberto Eco has given voice with "The Name of the Rose." What is significant about this comparison is how our era is populated with monsters just as the late Middle Ages was, which saw an equally intense and disruptive change as that which some see happening today. The cyborg, as Yehya reminds us, is at the center of this entire cultural process since, starting from the 1960s, the term "cyborg" has entered popular language and culture. Instead of colonizing galaxies, the cyborg has conquered domestic space, becoming the hero and villain of adventures of all kinds set in the future, influencing numerous military and civilian scientists, who have begun to shape a post-industrial and transhuman future. We must agree with Tagliasco when, looking at science fiction narratives, he asserts that in pop culture since the 1930s, science fiction has taken the place of mythology, religion, and philosophy in the role of transmitting myths and dreams. It will be only science fiction that will surreptitiously assume the task of keeping ancient dreams alive, immersing them in the anticipation of new technologies.

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Even if Tagliasco's conclusions about the role of science fiction may not be fully shareable, it remains true that science fiction production has seen a remarkable output and that it is certainly possible to trace in the pages of Sci-Fi the expectations, doubts, and perplexities that bind man to technology. In light of all this, we can try to outline some lines of horizon.

Economic development

Synthetic reality is the result of chance, the wrong outcome of some chemistry experiments in a London high school in the mid-nineteenth century. The fruit of a serendipity that, however, has been able to catalyze and attract Western industrial production, first producing synthetic dyes, then the first synthetic drugs, then plastics, and finally foods and living organisms. The synthetic is born at the intersection of various phenomena: industrial chemistry, the capitalist economy of the late nineteenth century, the military and state interests that marked the conflicts of the twentieth century, and mass culture. Synthetic reality first revealed itself to man as the possibility of creating substances and then became...
global and pervasive. The number of technical possibilities available to humanity raises the issue of the relationship that must exist between technical possibility and the moral possibility or legality of applying these same techniques. Some advocate a sort of ethical neutrality in the world of scientific research and technological innovation, claiming for the researcher every form of autonomy or moral freedom, asserting that, as a researcher, they are subject only to the scientific laws of the field in which they operate and not to moral laws, which would limit their operational space, also precluding the possibility of achieving results and new discoveries. This view excludes this sector of human activity from the realm of moral evaluation: explicitly or implicitly, this outcome is the result of overturning the hierarchical relationship between the value of scientific research and that of the beings affected by the research, especially when it comes to human beings, on whom potential harm falls or who are sacrificed in the name of the value of the research itself.

Upon closer examination, therefore, there is no ethical neutrality for an action of research or scientific experimentation, especially for the complex world created by the metaverse: this is aimed at achieving certain goals, involves other beings, and has consequences for them. However, noting a status of non-ethical neutrality does not automatically imply a negative moral judgment on it. Such activity can be morally acceptable or morally reprehensible: the task of ethical reflection is precisely the elaboration of such a judgment. In fact, the problem of the relationship between technical possibility and moral possibility is simply an aspect of the multiple ethical-normative problems of human activity. In particular, for our field of investigation, this can be formulated as follows: is the realization of new synthetic experiences between reality and digital, made possible by the metaverse, morally permissible or illicit?

In the next twenty years, the generation of children born in the third millennium will face three fundamental questions that the synthetic reality, now showing all its pervasiveness, presents. The resolution of these questions will describe, for better or worse, a world so profoundly different from anything humanity has yet experienced that we can truly imagine the end of an era. In light of this, I feel compelled to break with this recent sociological tradition of classifying young people with generational labels X, Y, and Z. It seems to me that due to the quality and characteristics of the technological change we have mediated in society, we should consider these kids—not these young people, that is, these aliens to a world made of natural and artificial—as an Omega Generation.

If we consider the philosophical, ethical, and practical challenges that synthetic reality presents and that these kids must face to build their identity as adult human beings, I think we can agree that this generation could be the last (human) generation—I am aware that the expression is strong and provocative, but I hope in the following pages to do justice to this provocation. The key theme is whether this generation will succeed in colonizing and urbanizing this new continent made of synthetic reality, mythical desires, and virtually unlimited technological potential. The ability of this generation could transform itself into something very different from what we currently understand as human. This is the generation that will have to take seriously the provocations of the synthetic and the posthuman—a complex and varied ecosystem of ideas that is forming around the emerging awareness that the traditional view of what constitutes a human being is subject to profound transformation.

If Hannah Arendt in the mid-twentieth century spoke of the human condition as the sum of the activities and capacities of man that constitute essential characteristics of human existence, it is now deemed necessary to speak of a posthuman condition that cannot be easily defined but is the condition of existence in which we find ourselves since the beginning of the posthuman era. The posthuman movement starts from the assumption that a profound transformation in human living has already occurred and that the result of this transformation generates a change in its way of being, ushering in the posthuman era. From this perspective, the posthuman movement, despite its heterogeneity and its...
Diversity, it differs from the numerous other futurist movements: those who identify as belonging to the posthuman current do not look to a possible future but to the present reality, recognizing that a radical change in the way of being human has already occurred. What we know is that the figure of the human who will inhabit our future is that of a wandering being in search. If it can respond to a spiritual call, it will return to being a viator; otherwise, it will self-condemn to being wandering and aimless. In fact, Generation Omega must respond, in a way that can no longer be postponed, to some fundamental questions about our human nature. These questions concern:

- **the relationship of humanity with its environment**
- **the relationship of humanity with technology**
- **the relationship of humanity with itself**

Let us briefly explore each of these questions.

a. The relationship of humanity with the environment
Stewart Brand created a slogan from the Whole Earth Catalog of 1968: “We are as gods and might as well get good at it.” The message was clear: human beings have reached a level of power and global impact on the Earth. As we have already seen, we have created a new geological era: the Anthropocene. If we want to survive, we must learn how to take responsibility for our entire global environment. From ocean acidification to soil depletion, from melting ice to dramatic changes in the chemical composition of our environment, the impact of population and the power of human technologization is decisive. This is a challenge that has no precedent in the entirety of global history - and it is a challenge that will fall squarely on the shoulders of Generation Omega. The resolution of this challenge will require profound systemic changes. For example, we will need to develop global collaborative capacities to seek solutions and strategies that are up to the task of ensuring the survival

of eight billion people endowed with extraordinary technological powers. This means much more than reaching a consensus on how the world works and how our actions affect it. It means learning to truly coordinate in a way that we have never experienced as human beings since, 70,000 years ago, we colonized the entire planet from Africa. In a vast global uncertainty, we will have to engage in large-scale geoengineering operations while pursuing intelligent and effective behaviors capable of managing the daily lives of every individual. It seems implausible. A utopian vision. Perhaps. But a utopia that for the first time will be built not out of aspiration, but out of necessity. As the aforementioned Stewart said, “We are as gods. We must become good.” The metaverse in this sense shows all its dual nature: it could be a tool for global collaboration or a highly effective means of mass alienation.

b. The relationship of humanity with technology
Perhaps the most astonishing truth of the modern age is that certain types of technology advance not along linear lines, but along exponential curves. Moore's famous law applies to batteries and bandwidth just as much as to processors. Every year, an ever-increasing portion of the world of technology is sucked into these exponential curves. Broadly speaking, this means that each year sees more "innovation" than all the previous years combined. The synthetic reality we have presented and reconstructed in its spread exemplifies this. This implies that the next twenty years will present changes so profound that they will render almost everything that has come before irrelevant. Science fiction and posthumanist communities have long been stirred and stimulated by the consequences of exponential technological growth. For Generation Omega, these speculations will firmly move into the realm of reality. Estimating these types of changes is notoriously difficult. Mathematically, with all the theoretical limits of the matter, if our technological capacity continues to grow at the same pace it has grown, in twenty years there will be at least a million people technologically more capable than we are today. A million times - in a single generation. It is a bit like going from the invention of writing to the invention of the computer - in a single generation. Humans as we currently know them have absolutely no idea how to adapt to such speed and scope of change. Let’s forget about self-driving cars, 3D printers, and autonomous drones. What will we be capable of doing? Certainly enhanced intelligence from cybernetics and detailed control over the genetic material of children with techniques like CRISPR. Probably technologies similar to telepathy and “swarm consciousness” where it becomes impossible to distinguish one’s own thoughts from the thoughts of the people one is connected to. Perhaps we will produce a virtual reality similar to The Matrix that is indistinguishable from reality. And perhaps digital superintelligence, which is the preferred dream of some posthumanists. Elon Musk continues to say: “I hope our generation is not just the bootloader.”
biological for digital super-intelligence.
Unfortunately, this is becoming increasingly likely.” Is the Singularity near?
Perhaps not yet, but it seems more and more probable that the Omega Generation will discover it. Regardless, it is likely that the Omega Generation will find itself exercising power over all various aspects of life much greater than what has so far been possible for humanity. The way we will manage such power is only imagined. But what awaits us could very well be more different from what we are today than we are from our hominid ancestors. Thanks to the metaverse, we could build new city-states for a strengthened global order based on the rule of law or new control bodies destined to be managed by a few private entities.

c. Humanity's relationship with itself
At this point, a crucial final question arises: how will humanity come to have the collective and individual wisdom to accept this responsibility and to exercise this technological power?
For those who are students of cultural anthropology, this question may be the most daunting. For millennia, we have - at least apparently - aspired to a world characterized by inner and outer peace. Great masters, philosophers, and leaders have walked among us, numerous great traditions have attempted to provide practices to bring us wisdom. Yet war, violence, and hatred are still a dominant part of our world. It seems a desperately foolish hope to think that in a single generation we could bring a critical mass of humanity to a level of wisdom, compassion, and integrity that is adequate for the task. Sometimes they have succeeded better, sometimes worse. Always we have found fragile and

precarious solutions.

This will also be the task of the Omega Generation. Until now, youth generations have been a national-level phenomenon. A generation is defined by a shared set of cultural sensibilities. According to an ever-increasing number of sociologists, we are witnessing the slow birth of a true global culture. Over the past seventy years, global media, global technology, global trade, and an increasing synchronization of global crisis events have only contributed to intensifying global culture. This means that the Omega Generation may not simply be the next generation of a country, but the next, and first, cross-border generation on the world stage. And, given the intrinsically global nature of their generational challenges, they are likely to be the first truly global generation.
There is no doubt that we are living in a moment characterized by strong unease. First, we must tear this sense of unease from the collective subconscious in which it resides. We must bring to light the causes that confine us within a restless horizon. The loss of hope, of tomorrow, and of a sense of a future are the result of a loss of the sense of the human. Only if we become aware of this loss of horizon can we find an answer. Secondly, we must return to being communities. If man is reduced to an individual, substituting all his relationships with technology, this atomized existence becomes isolated and lonely. But the sense, the capacity of man to go beyond the mere materiality of events, appears and becomes evident only in interpersonal relationships. Only by returning to a world of real and not virtual relationships will we be able to make those horizons of meaning reappear that will open a shared desire for tomorrow. Which, in other words, will make visible again that tension towards tomorrow that belongs to our human constitution and which shines fully in the Christian revelation.

This future belongs to the Omega Generation. Win or lose, theirs will be a generational bridge to an uncertain future. And there is no guarantee that they can successfully face these challenges. In fact, in all honesty, the odds are firmly against them. As we are now, especially with the recent international tensions growing daily, there are many reasons to fear and only a few reasons to hope.
But there are reasons to hope.
First among them is that the Omega Generation has not yet been educated. The oldest among them are not yet teenagers, and the youngest have not yet been born. They are still in the process of becoming who they will be, and therefore, we have the opportunity to give them the best possible chance while we, as adults, still hold the reins of power and can decide what world to leave them and what direction to guide them towards.
We know what they will have to face. What can we do now to help them?

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CONCLUSIONS
Ginevra Cerrina Feroni
Vice President of the Authority for the protection of personal data
CONCLUSIONS

Philosophies of the Metaverse between metaphysical, epistemological, and ethical issues

Ginevra Cerrina Feroni

Drawing conclusions from a conference like this is a difficult task, both due to the richness of the topic and the way this very topic has been addressed by the speakers. I will attempt to gather the insights that emerged from these rich presentations, focusing on three main trajectories. The first concerns the nature of immersive realities as such, that is, their essence; the second trajectory concentrates on the perspectives of these realities, including implications for privacy and personal identity of the participants; the third relates to the challenges arising from private governance. I will then leave to the conclusions (of the conclusions) reflections on the "why," that is, the reasons that drive individuals or entire societies to enter the metaverse.

As has been noted, the term "Metaverse" has noble, literary origins and is used to describe a shared collective virtual space, created by the convergence of immersive virtual reality and virtually enhanced physical reality. It is a happy lexical choice, striking in that the Greek preposition "meta," meaning "after," "beyond," perfectly fits the perspective of promoting the idea of a parallel world, that is, a metaphysics of the real. The Metaverse, at the moment the term has come to prominence in reference to a specific commercial operation, that of Meta, is now also considered a sort of metonymy for immersive realities in general. But rather than speaking of the Metaverse, we should talk about "metaverses" or simply immersive realities with some peculiar characteristics that will be examined shortly.

The Metaverse is, in the vision of its founder at Meta, "embodied Internet," where content can be experienced rather than merely viewed. The hypothesis of reality as simulation is ancient: the idea that our world and everything it contains are illusions has been widely addressed in literature—Calderón de la Barca posited that life is a dream (and some moments of our lives are a dream of a dream)—while for others, the world is already part of a simulation managed by some superior entity: for Hindus, we live in the dream of a God. There are also those who have drawn an intriguing parallel with Plato's Allegory of the Cave, to emphasize the illusory potentials that new technologies possess in relation to oneself and the world.

The possibility of voluntarily creating and accessing "virtual realities," illusory worlds in which the operator-observer becomes an actor, experiencing them as if they were real, is a hypothesis that the philosophy of science began to investigate more systematically only in the early 1990s. Numerous authors have captured its profound challenge regarding the meaning of reality and the relationship between reality and its representations. With "Real and Virtual," for example, Tomas Maldonado delved into one of the most fascinating, yet also most controversial, themes of contemporary cultural discourse: that of advanced technologies and their implications for life and culture in our time.

Perhaps the best perspective to understand what is happening is constituted by what are presented as the fundamental elements of the Metaverse, what must "build" it: virtual reality—or even better "immaterial," as Luciano Violante calls it in his compelling presentation—which allows for the exploration of new worlds and shared experiences; augmented reality, which enhances shared experiences with virtual effects; finally, wearable technology, which will become the true portal of access to the Metaverse, allowing interaction with the surrounding world.

If I had to think of a title that simply represents the enormous power of the digital, I might choose that of a book by Cosimo Accoto, "The World in Summary." In it, the ability to synthetically recreate the world is well expressed, just as one does with aromas that seem to provide the same flavor or transfer the same scent but are a chemical reproduction. "Clear and continuous are now the signs that herald the era of simulation," warns Accoto, who lists: artificial faces and cultivated meats, digital twins and virtual currencies, biosynthetic creatures and saturating metaverses, culminating in quantum machines and neuroprosthetic implants. "Simulation is a new terraformation," he concludes, describing the ongoing work in the global construction site of a new possible world. Essentially, what Slavoj Žižek had argued for robotization in "The Matrix" now applies to the entire world: a simulation of human life that is no longer just of bodies, but also of what surrounds them and the environment they inhabit.

This new context requires us to combine pragmatism and vision, that is, the ability to balance what is useful (governance, compliance, profit, etc.) with what has existential and immaterial meaning. But the answers regarding the meaning and reason of things can...
to arrive only from a new philosophy of the world, which is a synthetic world. It is no coincidence that already towards the end of the 1980s, IBM also hired graduates in philosophy, as logic and ethics are

at the core of the implications of computer science and its most futuristic applications. One thing is certain: understanding the line of development of the space, physical or digital, that surrounds us is not possible through traditional frameworks; the immaterial reality completely changes the perspective of existence as it impacts the brain (on mirror neurons, GPS neurons, attention networks, and interbrain neuronal oscillations), as Giuseppe Riva recalls in his speech and Maura Gancitano in her book "The New Gods."

However, understanding the line of development of the space, physical or digital, that surrounds us is not possible through traditional frameworks. "If we have stopped understanding the world, then it is precisely because the world has been remade, I would say, ontologically," writes Accoto, to signify that in the face of synthetic flesh, images created by algorithms, the Metaverse, genetic experimentation, and the ventures of the space economy, it is therefore natural to ask: what is flesh, what is face, what is space, what is life, what is world?

In the Metaverse, there will be protection of personal data, not privacy. I believe that the Metaverse has as its first effect that of impacting languages. A sort of semantic palingenesis, because if it seems complicated in itself to talk about data protection in the Metaverse, it seems more than an oxymoron to uphold the concept of privacy in meta-reality. Obviously, the utopian - and somewhat dystopian - perspective we think of concerns the integral replacement of our world with a new world. This replacement, as Agostino Ghiglia has also noted, will not happen immediately.

To access this Metaverse, it will be necessary, intuitively, to give up that specific hardware that today allows for concrete access: namely, the physical body. Scientific theories relaunched for the general public also by suggestive docuseries (I think of the episode "Earth" in Alien Worlds on Netflix) have already predicted the overcoming of our physicality: our existences (practically immortal) could evolve to the point of no longer needing a body to exist except in the form of mere neural tissue. Our physical body would be limited to cerebral matter protected and nourished in containment cells by artificial intelligence systems. Individual minds would be part of a single collective intelligence, and all our life, our sensations could be experienced in an entirely digital and highly interconnected reality. A very strong perspective.

At that point, we would have to ask ourselves what the true reality is, whether it is the physical (but vegetative) or the simulated (but intellectually vital). But to do this, a full interaction between physical stimuli and digital life is first necessary, so that the brain - as happens in the Turing test for AIs - can no longer distinguish whether we are facing a simulation or a reality. On the other hand, as far-fetched as it may seem, finding a way to exist in a world of data means, in essence, also increasing the life possibilities of individuals, and not only that: perhaps hunger can be eliminated, social and economic disparities can be eliminated, one can truly think of a State whose moral principles are regulated in a better way.

But without pushing so far into the future, we must at least think about the transition: mostly inserted into a new world, relocating the activities of the current world into that world, how these activities will need to be regulated. Are new rights (and duties) to be constructed solely for the Metaverses? Who will create the rules? Who will enforce them? Thinking hyperbolically, are personal data protection authorities conceivable in the Metaverse?

From this perspective, the risks of immersive digital realities are real: for example, there is the suppression or decrease of individuals' capacity for self-determination, to the advantage of formatting to a single dominant standard for the construction of further markets. The prevalence of virtual reality could easily lead to a new silent assault on human vulnerability exposed to the behavioral modulation of users/consumers, opening up various issues of social justice, commodification of personal data, and predatory practices on what is most precious we have as human beings: thoughts, emotions, consciousness.

Even today, in the face of "limited" Metaverses, it remains essential to guarantee the use of the most up-to-date security measures and a certain methodology in the management of data breaches, considering also that immersive virtual reality could generate new sources of illicit activities, just as the separation of a public dimension from a private one becomes indispensable, without this meaning anonymity.

Here, I believe that on this enormous topic, it is not...
Thinkable
to let the market take over. For this reason, the protection of personal data is no longer an “external” problem to the Metaverse, a simple regulation—one of many—to which to comply, nor is it sufficient to apply a simple “brushstroke” here and there for compliance purposes, just to say that one respects the supervisory authorities. Instead, the protection of personal data, especially in the context of a radical evolution like the one we are discussing, must be central to technological development and must accompany the architecture of the Metaverses from the very early stages of conception, design, and programming. This architecture must also be shaped based on the minimization of personal data processed and not modified ex post to recalibrate a project just enough to meet the existing privacy regulatory requirements.

That said, we must not overlook the issues that the digital reality brings with it. The real problematic issue to address will concern the relationship between content and support, that is, between the common good that will represent private property and the support on which this reality is based. Will we perhaps return to a sort of patrimonial state of feudal style in which the owners oversee and regulate our interaction by establishing the rules and their application? A patrimonial-public order where there is total coincidence between the person of the Lord (the big tech) and the private property of the land (the digital space where the Metaverse is built)?

In fact, there are analogies between the unequal contractual nature of the relationship between the feudal lord and the peasant community and the relationship between platform and user. To date, the Metaverses continue in the direction of reducing public space in favor of the private, even though the latter is called to publicize itself. However, I am convinced that the experiment of the functional Metaverse (and not just the playful one), if it does not turn out to be a flash in the pan, will require over time a nationalization or, at the very least, a significant democratization, as Guido Scorza also hinted in his speech, emphasizing the need for its “sustainable” development in democratic terms.

In such a scenario, the problem that arises is no longer how to access the personal data of the inhabitants of this reality, but only how to manage their processing. Already today, their owners acquire immense types of personal data: movements, emotions, psychological states, reactions, and every type of biometric data are extrapolated. A universal Metaverse, where normal daily activities are conducted, provides an enormous chance for manipulation of our social interactions, supported by the continuous monitoring of our consumption habits, our opinions, and our tastes, with the possible consequence that human behaviors soon become merchandise and that the integrity of personal identity is violated.

And here I am at the conclusions, or rather, at an attempt at a partial and not definitive conclusion. The last reflection I pose to myself is: why move to the Metaverses instead of observing the other places that make up our world? We have already spoken of the opportunities for enhanced life, and the increased possibility of interconnection, as well as the immense possibility of archiving our “self,” that identity we choose to share. On this point, it will be interesting to see the representation we will give of ourselves in the Metaverse. These would already be good and sufficient reasons to explain the “why” of the immersive digital reality.

However, there is another “why” of the Metaverse, much more seductive: the possibility of a planned life. The earthly order is constituted by the “things of the world” that Hannah Arendt speaks of, which have the task of stabilizing human life by offering it a foothold. Already today, and even more so with the implementation of the Metaverse, the earthly order is supplanted by the digital order. The digital order de-realizes the world, informatizing it. The new reality of the Metaverse will therefore be composed not of “things,” but of “non-things,” and this will profoundly alter the facts, that is, the reality as we perceive it. Already a few decades ago, media theorist Vilém Flusser observed: “Non-things are penetrating our environment from all directions, and driving out things. These non-things are called

information” or data. Byung-Chul Han writes in this regard that “with...
Digitalization: We Abandon the Terrestrial Order. Today
We no longer inhabit the sky and the earth, but rather the Cloud and Google Earth. To date, it is precisely the experience of presence that gives us the world, while digitalization leads to a poverty of presence and therefore of the world, to the point of no longer perceiving it except through its information, its data. Digitalization deprives us of the experience of presence.
But then, if we perceive the world only in terms of data, we remove the very premise for living reality: things. Things are fixed points of existence.
What can we say about data instead? Can they also serve as fixed points of existence? I have some doubts, if only because they have very limited validity and change continuously.
The tsunami of information that Bauman evoked, which has become credible in the Metaverse, even unsettles the cognitive system. In this sense, the Metaverse is not a stable construct: it definitively lacks the salvation of being. To put it in the words of the Korean-German philosopher, it is a “non-thing.”
One of the advantages - but also the psychological risks - of the Metaverses lies in programming, in being able, to a certain extent, to program one’s own experiences and in knowing that your experiences are programmed. If one’s imagination is found to be impoverished, one can utilize the library of suggestions drawn from biographies and enriched by novelists and psychologists. And then, just like literature, virtual places, in essence, soothe the anxiety of comparison, anticipating it, making us programmed and thus, perhaps, finally, real.
Here - and I admit all my limitations - personally, I am not able to say, and even before that, to understand, whether I like this perspective.
Thank you all.